

Elijah Valverde

Contact Information

email: evalverde@sfsu.edu
website: elijahvalverde.github.io

Education

San Francisco State University
Bachelor of Arts, Mathematics 2022-
GPA: 4.0

Preprints

A Sign-Reversing Involution on Genus-One Boundary Strata, with Emily Clader

Research Experience

Vertex Cones of Order Polytopes Aug 2025-
Studying the combinatorics and integer-point transforms of vertex cones of an order polytope.
Advised by Matthias Beck and Emily Clader, partially supported by CSU Chancellor's Office Grant No. HRD-2308501

Sign-Reversing Involution on Genus-One Boundary Strata Spring 2025
We give an elementary calculation of the Euler characteristic of the boundary complex of the moduli space of genus-one stable curves via a sign-reversing involution.
Advised by Emily Clader, partially supported by NSF Grant No. HRD-1826490 and NSF CAREER Grant 2137060

Gaussian Processes Applied to High-Time Aerosol Datasets Summer 2024
Applied a Gaussian Process to aerosol datasets collected during the Colorado SAIL campaign to aid researchers with future aerosol predictions.
Advised by Allison Aiken, partially supported by Department of Energy (DOE) Science Undergraduate Laboratory Internship (SULI) program

Awards

Highlighted Awards
(\$30,000) 2025 CSU-UC Cal-Bridge Fellowship, 50 recipients across Math, CS and Physics & Astronomy majors at all 22 CSU's
(\$5,000) 2025 CSU Predoctoral Sally Casanova Fellowship, 80 recipients across all majors at 22 CSU's
(\$12,000) 2024 Department of Energy Science Undergraduate Laboratory Internships Scholar, 1000 recipients across United States
(\$1,500) 2024 Halmos Scholarship, 3 recipients in mathematics department at SFSU

2025

Math Matters Scholarship, *San Francisco State University*
Pamela Fong Scholarship, *San Francisco State University*
LSAMP Scholar, *San Francisco State University*
Deans List 1x, *San Francisco State University*

2024

Pamela Fong Scholarship, *San Francisco State University*
C.Y. Chow Memorial Scholarship, *San Francisco State University*
Best Mathematics Poster, *Los Alamos Student Symposium*
Deans List 2x, *San Francisco State University*

≤ 2023

San Francisco State University Scholarship, *San Francisco State University*
Vincent Constantino Scholarship, *San Francisco State University*
SIAM Student Chapter Certificate of Recognition 2021–2022, *SIAM*
Deans List 4x, *San Francisco State University*

Presentations	<i>Joint Mathematics Meetings</i>	Jan 2026
	<i>Cal-Bridge Student Research Symposium</i>	Oct 2025
	<i>SFSU SIAM Student Symposium</i>	Jan 2026
	<i>SFSU College of Science and Engineering Student Project Showcase</i>	May 2025
	<i>Joint Mathematics Meetings</i>	Jan 2025
	<i>Northern and Central California Section of SIAM Annual Meeting</i>	Oct 2024
	Honorable mention for best undergraduate poster.	
	<i>Los Alamos National Laboratory Student Research Symposium</i>	July 2024
	Winner of best mathematics poster.	
Community Engagement	Mathematistas, Member	Fall 2025
	Aid in coordinating events for mathematistas, a club at San Francisco State striving towards gender equity in mathematics.	
	Mathematics Department, Teaching Assistant	Jan 2024 -
	Worked as a teaching assistant for Analysis I and Abstract Algebra I & II. Services include grading, helping students with course questions and lecturing (sparingly).	
	Society for Industrial and Applied Mathematics, President	Aug 2023 -
	Currently serving as president since Aug 2025, held previous positions as Vice President and Officer. In collaboration with other officers, I coordinated research symposiums, career development events, and hosted talks from applied mathematicians in Industry and Academia.	
	Tutoring Center San Francisco State University, Tutor	Jan 2023-
	I offer tutoring services for courses ranging from Precalculus to graduate Algebra. Also served as a teaching assistant embedded in classrooms with high DFW (Drop, Fail or Withdraw) rates.	